

HRIDAYAM KAPILA

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PROFILE

M1 student in Applied Economics at Toulouse School of Economics and economics graduate of Aryabhata College, University of Delhi. Targeting pre-doctoral research positions and PhD programs in environmental and resource economics and sustainable development, with interests in conservation, bioeconomics, and quantitative modeling.

EDUCATION

Toulouse School of Economics

2026–Present

M1 Applied Economics

Current coursework: Game Theory, Macroeconomics, Intermediate Econometrics, R Programming, Environmental Economics, Markov Chains.

Aryabhata College, University of Delhi

2022–2026

B.A. (Honours with Research) in Economics, Minor in Mathematics

Economics: Intermediate Microeconomics and Macroeconomics I and II, Environmental Economics, Optimization Methods for Economic Analysis, Basic Econometrics, Advanced Econometrics, Introduction to Causal Inference, Forecasting Methods for Economics.

Mathematics: Topics in Calculus, Elementary Linear Algebra, Differential Equations, Abstract Algebra, Real Analysis, Probability and Statistics.

RESEARCH INTERESTS

Environmental Economics, Ecological Economics, Bioeconomics, Conservation, Climate Change, Applied Econometrics, Economic Geography.

PAPERS UNDER REVIEW

- Regional Integration and Carbon Emissions in Africa: The Role of Value Chains, Environmental Cooperation, and Institutional Integration. Under review at *Environment Systems and Decisions* (2026).

RESEARCH EXPERIENCE

Undergraduate Thesis

Aug 2025 – May 2026

Carbon Emissions in Africa: EKC and Regional Integration

Supervisor: Prof. Deepika Goel

The thesis studies the long-run relationship between carbon emissions and regional integration in a panel of nine African countries (Algeria, Kenya, Mauritius, Morocco, Mozambique, Namibia, Nigeria, South Africa, and Zambia), 2006–2022.

Paper 1 measures integration through regional value chains, environmental cooperation, and institutional ties. Using fully modified OLS to account for cross-sectional dependence, non-stationarity, and endogeneity, it finds the variables cointegrated: trade openness and regional value chains are positively associated with consumption-based emissions, while environmental cooperation and institutional integration are negatively associated with them.

Paper 2 tests for an Environmental Kuznets Curve with a group-mean fully modified OLS estimator. It finds a long-run cointegrating relationship with heterogeneous slopes and a U-shaped income–emissions pattern instead of the conventional inverted U, with an almost one-to-one elasticity between energy use and emissions per capita.

PRESENTATIONS

Carbon Emissions in Africa: EKC and Regional Integration

April 2026

Undergraduate Research and Entrepreneurship Exhibition (UGREE) 2026, University of Delhi.

PROJECTS

Cartel-Fringe Resource Model (Julia/JuMP) ([GitHub](#))

Built a T-period resource-extraction model in the Salant (1976)/Hotelling (1931) tradition using Julia/JuMP, with a complementarity formulation for fringe supply, and simulated how alternative carbon-tax paths affect extraction timing.

INR/USD Macroeconomic Analysis: OLS and VAR (R) ([OLS](#) | [VAR](#))

Analyzed monthly INR/USD movements, inflation, and industrial activity in R. Rebuilt the levels OLS (127 observations) as a stationary specification after unit-root diagnostics, and extended the work into a VAR of exchange-rate and inflation changes with the IIP growth cycle (75 observations). Used ADF/KPSS and bootstrap unit-root tests, ARDL bounds tests, block Granger tests, and Newey-West inference.

QS Rankings Analysis (Tableau) ([Tableau Public](#))

Interactive dashboard visualizing and analyzing university ranking data.

AWARDS AND HONORS

- Fee waiver for 2026–2027, Toulouse School of Economics (2026)
- Undergraduate thesis scored 10/10, top of the Economics department, University of Delhi (2026)

CERTIFICATIONS

- TOEFL iBT: 108/120 (C1), ETS (2025)
- Data Science Professional Certificate, Harvard Online (2024)
- CS50's Introduction to Programming with Python, HarvardX/edX (2024)
- Introduction to Data Science with Python, edX (2024)
- Learning Python for Data Science, Harvard Online (2024)
- Fat Chance: Probability from the Ground Up, edX (2024)
- Logic for Economists, University of Amsterdam (2024)
- Introduction to Economic Theories, Erasmus University Rotterdam (2023)
- SQL for Beginners, Udemy (2024)
- Statistics and Data Analysis using R, University of Delhi (2025)
- Fundamentals of Visualization with Tableau, UC Davis (2024)
- NISM Certification (Mutual Funds), National Institute of Securities Markets (2024)

SKILLS

Languages: Hindi (Native), English (Fluent), Punjabi (Elementary), French (Elementary)

Technical Skills: R, Stata, EViews, Tableau, LaTeX, Git/GitHub, Excel, Typst, Julia (JuMP), Numerical Optimization.

LEADERSHIP

Placement Unit, Department of Economics

Sep 2025 – Apr 2026

Aryabhatta College, University of Delhi

- **Co-Founder and Vice President:** Established and led the unit to connect economics students with internship and career opportunities.

Mind Over Matter (Economics Society)

Oct 2023 – Apr 2026

Aryabhatta College, University of Delhi

- **Advisory** (Aug 2025 – Apr 2026): Provided strategic guidance and mentorship to the editorial board.
- **Editor-in-Chief** (Oct 2024 – Aug 2025): Led the editorial team, managing content strategy, publication processes, and project communications.
- **Associate** (Oct 2023 – Oct 2024)

Arahata (B.A. Programme Society)

Oct 2023 – Jul 2025

Aryabhatta College, University of Delhi

- **Social Media Manager** (Nov 2024 – Jul 2025): Managed the society's online presence and communication strategies.
- **Treasurer** (Oct 2023 – Oct 2024): Managed society finances, including budget planning, fund allocation, and expense tracking.